

Ratio: sharing and missing amounts

Worked demo • Follow each step down your column.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A and B share £120 in the ratio 2:3. How much does B receive?	Red and blue paint are mixed in the ratio 3:5. There are 18 litres of red paint. How many litres of blue paint are needed?	Red and blue counters are in the ratio 3:7. There are 28 more blue counters than red counters. How many counters are there altogether?
Keywords & Calculation	“share £120”: the total is known. $2 + 3 = 5$ parts. $£120 \div 5$, then $\times 3$.	“18 litres of red”: 3 parts are known. $18 \div 3$, then $\times 5$.	“28 more blue”: the difference is known. $7 - 3 = 4$ parts. $28 \div 4$, then $\times (3 + 7)$.
Written Method	$£120 \div 5 = £24$ per part. $£24 \times 3 = £72$. A B £120 altogether 1 part = £24	$18 \div 3 = 6$ litres per part. $6 \times 5 = 30$ litres. Red Blue Red: 18 litres 1 part = 6 litres	$28 \div 4 = 7$ counters per part. $3 + 7 = 10$ parts. $7 \times 10 = 70$ counters. Red Blue 4 extra parts = 28 counters 1 part = 7 counters
Ballpark Check & Math	B gets more than half of £120, so more than £60. £72 fits. Check both shares: $£48 + £72 = £120$.	Blue is between one and two times the red amount, so between 18 and 36 litres. 30 litres fits.	The total represents 10 parts and the difference 4. The total should be between 2 and 3 times 28: between 56 and 84. 70 fits.
Final Answer	B receives £72.	30 litres of blue paint.	70 counters altogether.

Ratio: sharing and missing amounts

Step-by-step practice 1 • Complete each step in your booklet or webapp.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A and B share £84 in the ratio 3:4. How much does A receive?	Notebooks and folders are in the ratio 4:7. There are 35 folders. How many notebooks are there?	Adults and children are in the ratio 2:5. There are 24 more children than adults. How many people are there altogether?
Keywords & Calculation	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.
Written Method	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).
Ballpark Check & Math	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?
Final Answer	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.

Ratio: sharing and missing amounts

Step-by-step practice 2 • Complete each step in your booklet or webapp.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A and B share £96 in the ratio 5:7. How much does B receive?	Juice and water are mixed in the ratio 2:7. There are 18 litres of juice. How many litres of water are needed?	Two lengths of ribbon, A and B, are in the ratio 5:8. B is 21 cm longer than A. What is their total length?
Keywords & Calculation	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.
Written Method	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).
Ballpark Check & Math	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?
Final Answer	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.

Ratio: sharing and missing amounts

Independent practice 1 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A and B share £40 in the ratio 1:3. How much does A receive?	1 Red and blue counters are in the ratio 1:3. There are 10 red counters. How many blue counters are there?	1 A and B share money in the ratio 2:5. B receives £33 more than A. How much does A receive?
2 A and B share £60 in the ratio 2:3. How much does B receive?	2 Red and blue counters are in the ratio 2:3. There are 36 blue counters. How many red counters are there?	2 A and B share money in the ratio 3:7. B receives £52 more than A. How much does B receive?
3 A and B share £64 in the ratio 3:5. How much does A receive?	3 Red and blue counters are in the ratio 3:5. There are 24 red counters. How many counters are there altogether?	3 A and B share money in the ratio 4:9. B receives £45 more than A. Find the total amount.
4 A and B share £54 in the ratio 4:5. How much does B receive?	4 Red and blue counters are in the ratio 4:5. There are 30 blue counters. How many red counters are there?	4 A and B share money in the ratio 5:8. B receives £21 more than A. How much does A receive?
5 A and B share £81 in the ratio 2:7. How much does A receive?	5 Red and blue counters are in the ratio 2:7. There are 18 red counters. How many blue counters are there?	5 A and B share money in the ratio 7:3. A receives £40 more than B. How much does B receive?
6 A and B share £84 in the ratio 5:7. How much does B receive?	6 Red and blue counters are in the ratio 5:7. There are 49 blue counters. How many counters are there altogether?	6 A and B share money in the ratio 8:5. A receives £24 more than B. Find the total amount.

Ratio: sharing and missing amounts

Independent practice 2 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A and B share £84 in the ratio 2:5. How much does B receive?	1 Red and blue counters are in the ratio 2:5. There are 60 blue counters. How many red counters are there?	1 A and B share money in the ratio 3:5. B receives £26 more than A. How much does B receive?
2 A and B share £98 in the ratio 3:4. How much does A receive?	2 Red and blue counters are in the ratio 3:4. There are 42 red counters. How many blue counters are there?	2 A and B share money in the ratio 2:7. B receives £75 more than A. Find the total amount.
3 A and B share £110 in the ratio 5:6. How much does B receive?	3 Red and blue counters are in the ratio 5:6. There are 60 blue counters. How many counters are there altogether?	3 A and B share money in the ratio 5:9. B receives £44 more than A. How much does A receive?
4 A and B share £88 in the ratio 3:8. How much does A receive?	4 Red and blue counters are in the ratio 3:8. There are 24 red counters. How many blue counters are there?	4 A and B share money in the ratio 7:4. A receives £27 more than B. How much does B receive?
5 A and B share £99 in the ratio 7:2. How much does B receive?	5 Red and blue counters are in the ratio 7:2. There are 22 blue counters. How many red counters are there?	5 A and B share money in the ratio 9:5. A receives £48 more than B. Find the total amount.
6 A and B share £108 in the ratio 7:5. How much does A receive?	6 Red and blue counters are in the ratio 7:5. There are 63 red counters. How many counters are there altogether?	6 A and B share money in the ratio 8:3. A receives £50 more than B. How much does A receive?

Ratio: sharing and missing amounts

Independent practice 3 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A and B share £154 in the ratio 4:7. How much does A receive?	1 Red and blue counters are in the ratio 4:7. There are 56 red counters. How many blue counters are there?	1 A and B share money in the ratio 4:7. B receives £45 more than A. Find the total amount.
2 A and B share £128 in the ratio 5:3. How much does B receive?	2 Red and blue counters are in the ratio 5:3. There are 48 blue counters. How many red counters are there?	2 A and B share money in the ratio 3:8. B receives £85 more than A. How much does A receive?
3 A and B share £132 in the ratio 2:9. How much does A receive?	3 Red and blue counters are in the ratio 2:9. There are 24 red counters. How many counters are there altogether?	3 A and B share money in the ratio 7:9. B receives £26 more than A. How much does B receive?
4 A and B share £150 in the ratio 7:8. How much does B receive?	4 Red and blue counters are in the ratio 7:8. There are 80 blue counters. How many red counters are there?	4 A and B share money in the ratio 8:5. A receives £33 more than B. Find the total amount.
5 A and B share £169 in the ratio 9:4. How much does A receive?	5 Red and blue counters are in the ratio 9:4. There are 117 red counters. How many blue counters are there?	5 A and B share money in the ratio 9:4. A receives £70 more than B. How much does A receive?
6 A and B share £154 in the ratio 5:9. How much does B receive?	6 Red and blue counters are in the ratio 5:9. There are 99 blue counters. How many counters are there altogether?	6 A and B share money in the ratio 6:11. B receives £60 more than A. How much does B receive?

Ratio: sharing and missing amounts

Independent practice 4 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A and B share £160 in the ratio 3:7. How much does B receive?	1 Red and blue counters are in the ratio 3:7. There are 112 blue counters. How many red counters are there?	1 A and B share money in the ratio 5:7. B receives £34 more than A. How much does A receive?
2 A and B share £198 in the ratio 6:5. How much does A receive?	2 Red and blue counters are in the ratio 6:5. There are 108 red counters. How many blue counters are there?	2 A and B share money in the ratio 4:9. B receives £95 more than A. How much does B receive?
3 A and B share £182 in the ratio 4:9. How much does B receive?	3 Red and blue counters are in the ratio 4:9. There are 126 blue counters. How many counters are there altogether?	3 A and B share money in the ratio 8:11. B receives £45 more than A. Find the total amount.
4 A and B share £120 in the ratio 7:3. How much does A receive?	4 Red and blue counters are in the ratio 7:3. There are 84 red counters. How many blue counters are there?	4 A and B share money in the ratio 9:5. A receives £52 more than B. How much does A receive?
5 A and B share £195 in the ratio 8:5. How much does B receive?	5 Red and blue counters are in the ratio 8:5. There are 75 blue counters. How many red counters are there?	5 A and B share money in the ratio 11:7. A receives £64 more than B. How much does B receive?
6 A and B share £208 in the ratio 9:7. How much does A receive?	6 Red and blue counters are in the ratio 9:7. There are 117 red counters. How many counters are there altogether?	6 A and B share money in the ratio 12:5. A receives £98 more than B. Find the total amount.